

Table 1: Sensitivity to the Number of CHSW Projection Directions K . The best results are highlighted in bold.

Dataset	K	ACC $\uparrow$	$\Delta_{DP} \downarrow$	$\Delta_{DP}^{\text{soft}} \downarrow$
Pullback	60	0.819$\pm$0.019	0.293 $\pm$ 0.051	0.257 $\pm$ 0.057
	120	0.758 $\pm$ 0.064	0.187$\pm$0.122	0.176$\pm$0.096
	240	0.807 $\pm$ 0.040	0.268 $\pm$ 0.088	0.243 $\pm$ 0.075
Hyperbolic	60	0.485 $\pm$ 0.055	0.229$\pm$0.159	0.013 $\pm$ 0.009
	120	0.521$\pm$0.061	0.245 $\pm$ 0.158	0.005$\pm$0.006
	240	0.513 $\pm$ 0.073	0.281 $\pm$ 0.160	0.008 $\pm$ 0.006
SPD	60	0.785 $\pm$ 0.048	0.221 $\pm$ 0.130	0.088 $\pm$ 0.057
	120	0.795$\pm$0.034	0.210$\pm$0.145	0.079 $\pm$ 0.068
	240	0.793 $\pm$ 0.034	0.221 $\pm$ 0.137	0.078$\pm$0.066
Adult	60	0.843$\pm$0.005	0.115 $\pm$ 0.008	0.098 $\pm$ 0.017
	120	0.840 $\pm$ 0.007	0.098 $\pm$ 0.034	0.082 $\pm$ 0.033
	240	0.823 $\pm$ 0.040	0.081$\pm$0.054	0.067$\pm$0.048
LFWA	60	0.902 $\pm$ 0.011	0.059 $\pm$ 0.010	0.058 $\pm$ 0.009
	120	0.905$\pm$0.004	0.048$\pm$0.012	0.046$\pm$0.010
	240	0.902 $\pm$ 0.005	0.057 $\pm$ 0.009	0.051 $\pm$ 0.009

Table 2: Runtime (ms/epoch) and Memory (MB) comparisons. Runtime is averaged per epoch over 10 epochs; for WB (post-processing), we report its one-time cost. Memory denotes peak GPU usage during training.

Method	Pullback		Hyperbolic		SPD		Adult		LFWA	
	Runtime	Memory	Runtime	Memory	Runtime	Memory	Runtime	Memory	Runtime	Memory
M _{AVG}	41.9	0.8	240.0	0.7	17.1	18.6	765.3	1.6	1680.3	188.6
M _{MMD}	36.3	2.3	256.2	2.2	21.9	20.2	990.6	3.2	1673.7	188.6
M _{SNK}	420.7	2.3	3098.0	2.3	255.9	20.2	18115.7	22.9	4058.8	188.6
WB	25.6	0.8	160.8	0.7	18.2	18.6	483.7	1.6	1828.7	358.5
SW	43.0	2.2	297.9	2.2	26.2	19.2	920.3	3.1	2187.1	358.5
Ours	48.9	2.3	321.9	6.9	36.0	19.3	925.5	3.1	2095.7	188.6

Table 3: Sensitivity to the Choice of Origin o . The best results are highlighted in bold.

Dataset	Origin	Description	ACC $\uparrow$	$\Delta_{DP} \downarrow$	$\Delta_{DP}^{\text{soft}} \downarrow$
Hyperbolic	center ($\ o\ = 0$)	canonical origin (Poincaré ball center)	0.521$\pm$0.061	0.245 $\pm$ 0.158	0.005$\pm$0.006
	$\ o\ = 0.15$	random point with small displacement	0.502 $\pm$ 0.065	0.236 $\pm$ 0.153	0.009 $\pm$ 0.008
	$\ o\ = 0.30$	random point with larger displacement	0.477 $\pm$ 0.052	0.226$\pm$0.160	0.011 $\pm$ 0.008
	data_mean	heuristic: Fréchet mean of all samples	0.474 $\pm$ 0.052	0.245 $\pm$ 0.142	0.010 $\pm$ 0.008
	group_barycenter	adaptive: fairness-aware group barycenter	0.504 $\pm$ 0.064	0.234 $\pm$ 0.152	0.009 $\pm$ 0.008
SPD	identity (I)	canonical origin (identity matrix)	0.795$\pm$0.034	0.210 $\pm$ 0.145	0.079 $\pm$ 0.068
	random1	random SPD matrix (seed=42)	0.778 $\pm$ 0.044	0.188 $\pm$ 0.197	0.081 $\pm$ 0.087
	random2	random SPD matrix (seed=123)	0.793 $\pm$ 0.037	0.179$\pm$0.159	0.073$\pm$0.076
	data_mean	heuristic: Fréchet mean of all samples	0.789 $\pm$ 0.050	0.249 $\pm$ 0.167	0.120 $\pm$ 0.077
	group_mean	adaptive: group-wise barycenter	0.790 $\pm$ 0.047	0.263 $\pm$ 0.169	0.125 $\pm$ 0.082

Table 4: Comparison with an Adversarial FRL Baseline on Adult and LFWA. The best results are highlighted in bold.

Method	Adult			LFWA		
	ACC $\uparrow$	$\Delta_{DP} \downarrow$	$\Delta_{DP}^{\text{soft}} \downarrow$	ACC $\uparrow$	$\Delta_{DP} \downarrow$	$\Delta_{DP}^{\text{soft}} \downarrow$
CHSW	0.840$\pm$0.007	0.098$\pm$0.034	0.082$\pm$0.033	0.905$\pm$0.004	0.048$\pm$0.012	0.046$\pm$0.010
LAFTTR	0.835 $\pm$ 0.003	0.168 $\pm$ 0.009	0.176 $\pm$ 0.003	0.893 $\pm$ 0.004	0.074 $\pm$ 0.006	0.068 $\pm$ 0.004